



School of Engineering

Data-Driven Retail Insights for Cost Savings & Revenue Growth



[InstaCart Online Grocery Basket Analysis Dataset](#)

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Introduction

Retail grocery businesses operate in an environment where margins are thin and promotional spending is high. Despite the availability of transactional data, many retailers still rely on uniform discount strategies that ignore differences in customer behavior and product-level economic contribution.

The core problem addressed in this project is the following:

How can transactional basket data be used to increase revenue and reduce promotional inefficiency through targeted, data-driven strategies?

Using the Instacart Online Grocery Basket Analysis dataset, this project develops a structured analytical framework to:

- Segment customers based on purchasing behavior
- Identify stable co-purchase patterns
- Integrate pricing signals to evaluate economic impact
- Construct segment-specific product bundles

The goal is not merely to describe purchasing behavior, but to translate it into financially actionable retail strategy.

The business analyze : [POWER BI](#)

2. Dataset Description

This project uses the Instacart Online Grocery Basket Analysis dataset, which contains over three million grocery orders from more than 200,000 customers. The data includes order identifiers, customer IDs, product IDs, aisle and department classifications, and reorder indicators. This structure enables reconstruction of shopping baskets and longitudinal customer behavior.

The dataset supports analysis of purchase frequency, basket size, product loyalty, department penetration, and co-purchase relationships, making it suitable for both segmentation and association rule mining.

To increase business relevance, we enriched the transactional data with:

- **Pricing information:** unit price proxies used to approximate monetary contribution per basket.

- **Health indicators (NutriScore):** where available, used to estimate health orientation of purchases.

This enrichment allows the analysis to move beyond frequency-based patterns and introduce value-aware interpretation.

3. Methodology

3.1 Exploratory Data Analysis

Prior to model development, a deep-dive exploratory analysis was conducted using a high-performance data processing stack (Python, Polars, and Parquet) to handle the 33 million record granularity. The objective was to move beyond descriptive statistics to uncover the "Store Pulse" and the underlying mechanics of customer loyalty.

a. Dataset Scale and Intensity Distribution

The dataset comprises over 33.8 million order-product records from 206,000 users. Technical profiling revealed:

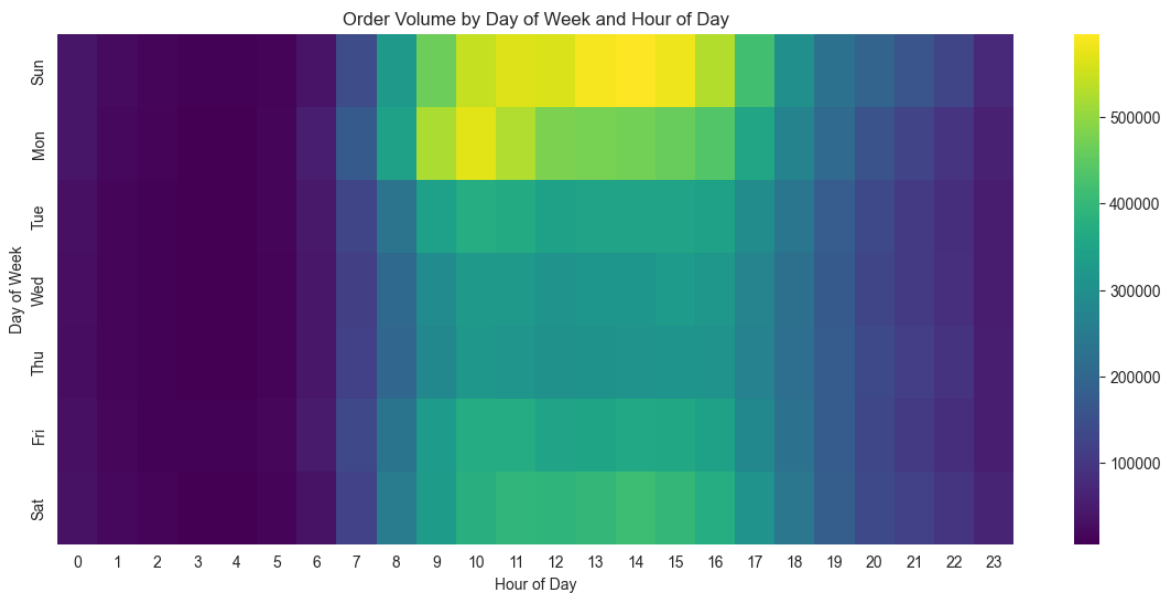
- **Shopping Intensity:** A median basket size of 8 items with an average of 10.2, but more importantly, a high coefficient of variation (CV) in order frequency. This statistical dispersion confirmed that a "one-size-fits-all" model would be suboptimal, justifying the subsequent unsupervised clustering approach.
- **Data Architecture:** To maintain computational efficiency, data was processed using Lazy Evaluation and categorical encoding for high-cardinality features (Product IDs, User IDs), ensuring sub-second query times on the full 33M-row corpus.

b. Temporal Purchasing Fingerprints (The "Store Pulse")

Instead of viewing time as a linear variable, we mapped Temporal Fingerprints—heatmaps crossing hour-of-day with day-of-week.

- **Key Finding:** Peak activity occurs between 10:00 AM and 3:00 PM, with a significant "Sunday-Monday surge."

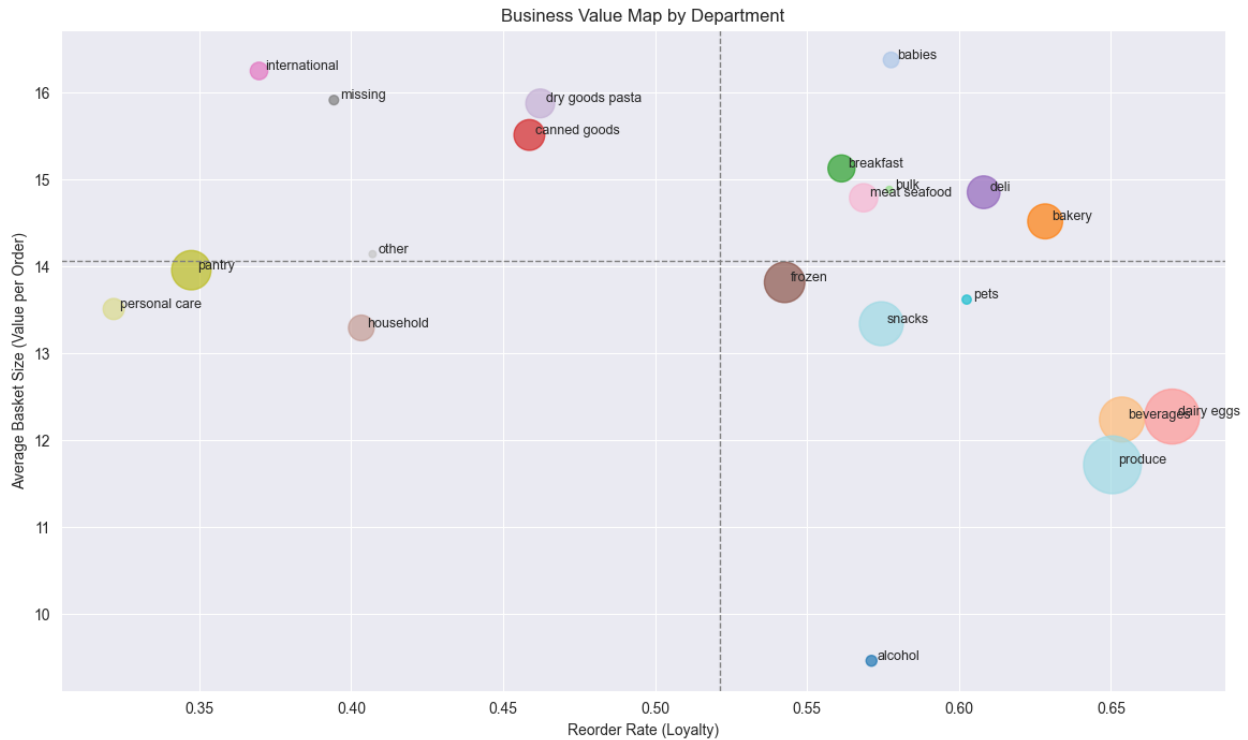
- Strategic Insight: Early morning orders (6:00 AM - 9:00 AM) exhibit a significantly higher Reorder Ratio, identifying a "Routine Staple" segment (Coffee, Breakfast) that differs from the more exploratory "Afternoon Discovery" shopping sessions.



c. The Loyalty Landscape & Value Mapping

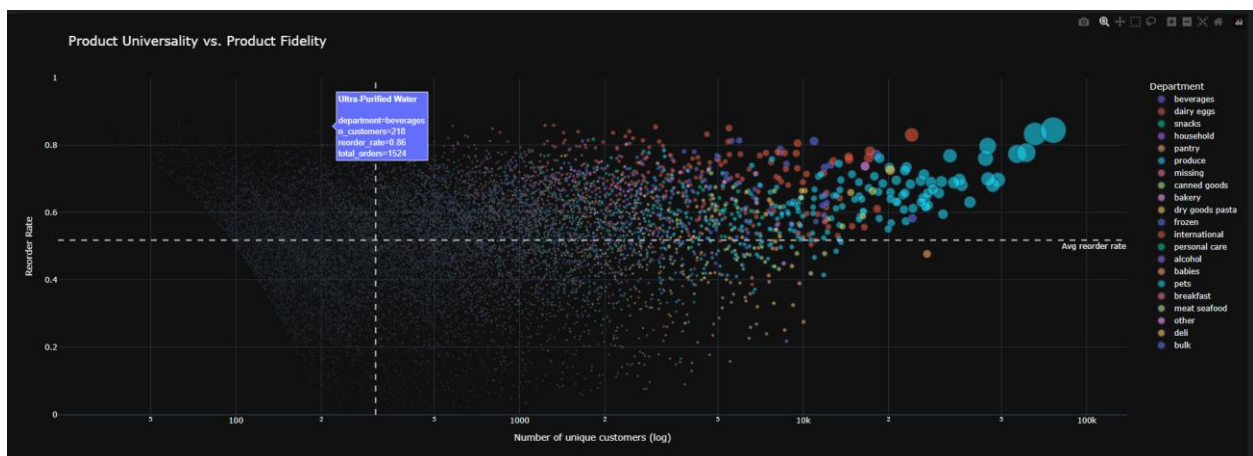
A multi-dimensional analysis of reorder dynamics was performed to distinguish between transactional volume and customer retention:

- Departmental Stratification: We identified that while *Produce* and *Dairy* drive the highest volume, specific "Niche" aisles (e.g., *Baby Food*, *Pet Care*) exhibit the highest Product Fidelity (reorder rates >75%).
- The Value Map (Quadrant Analysis): By crossing *Order Frequency* with *Average Basket Value*, we categorized departments into four strategic quadrants:
 1. Value Champions: High frequency, high basket size (Produce, Dairy).
 2. Loyal Niches: Low frequency but high unit value/fidelity (Organic Specialties).
 3. Popular Commodities: High frequency, low margin (Snacks, Beverages).
 4. Low Value/Churn Risk: Low engagement across all metrics.



d. Extraction of "Hidden Gems"

A technical filter was applied to identify products with low total support (volume) but extreme reorder rates. These "Hidden Gems" represent high-affinity anchors. Identifying these products was crucial for our Association Rule Mining, as it allowed us to weight rules not just on *Support* (frequency) but on *Lift* (predictive power).



e. Synthesis for Feature Engineering

These EDA observations directly informed our Feature Engineering Pipeline:

- **Indices Creation:** Development of the HAI (Healthy Affinity Index) and PSI (Price Sensitivity Index) based on the observed concentration in specific departments.
- **Validation Strategy:** The high variance in user behavior led to the implementation of a User-Level Stratified Split for the Association Rules, ensuring that our training set was behaviorally representative of our test set.

3.2 Association Rule Mining: What Do Customers Buy Together ?

To identify co-purchase patterns, we implemented a robust Association Rule Mining pipeline using the FP-Growth algorithm. Unlike traditional approaches that run rules on a global dataset, our methodology focused on model generalizability and statistical stability.

3.2.1. Behavioral Stratification & Sampling

To prevent sampling bias, we developed a **User-Level Stratification** before splitting the data:

- **Persona Creation:** Users were discretized into 9 distinct behavioral strata based on two axes: **Order Frequency** (days since prior order) and **Basket Volume** (average items per order).
- **Stratified Train/Test Split:** We performed an 80/20 split on User IDs (not rows), ensuring that the test set (simulating "new" customers) perfectly mirrored the behavioral distribution of the training set. This is a critical step to ensure that the rules discovered are not overfitted to a specific type of shopper.

3.2.2. Frequent Pattern Mining (FP-Growth)

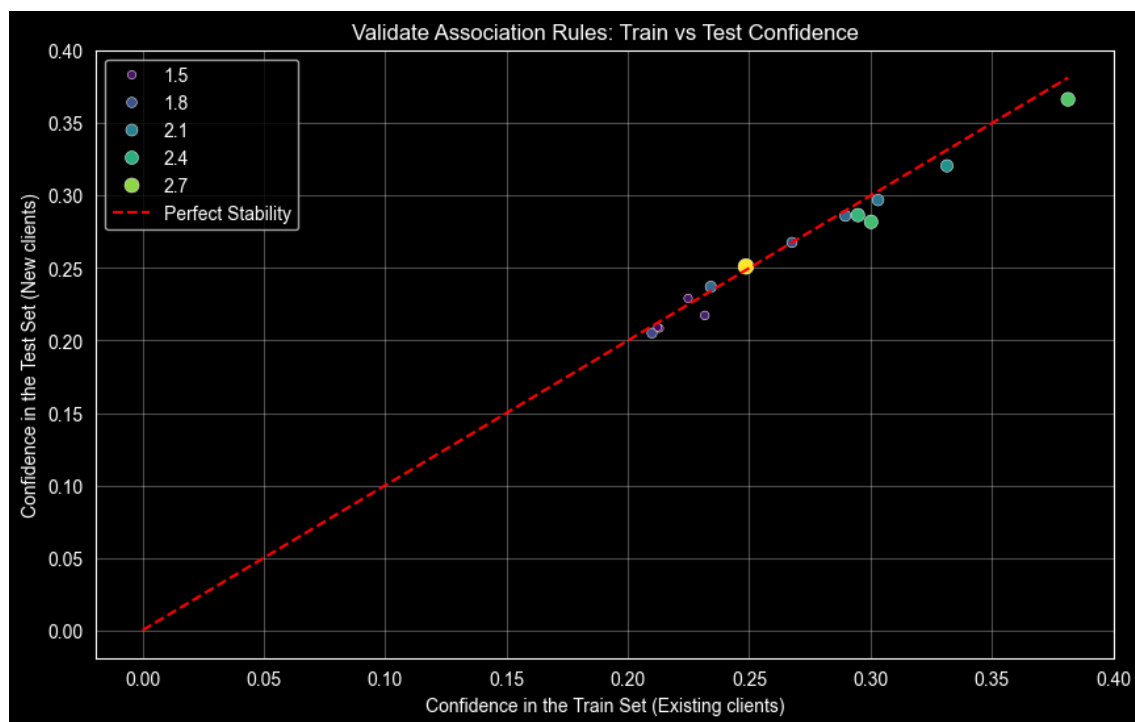
The mining phase was optimized for both computational efficiency and commercial relevance:

- **Dimensionality Reduction:** We applied a support threshold, focusing on products with >1,000 sales to filter out "long-tail" noise and ensure statistical significance.
- **Sparse Matrix Encoding:** To handle the 33M-record granularity, we utilized a Sparse CSR (Compressed Sparse Row) matrix format, drastically reducing the memory footprint during itemset generation.
- **Rule Extraction:** Frequent itemsets were generated with a minimum confidence of 20%, focusing on **Lift** as the primary metric to identify relationships that occur significantly more often than by random chance.

3.2.3. Cross-Population Stability Validation

The most innovative part of the pipeline is the Cross-Population Validation. Each rule generated on the training population was re-calculated on the unseen test population:

- **Confidence Gap Analysis:** We measured the "Stability Delta" (absolute difference between Train and Test confidence).
- **Stability Filtering:** Only rules that maintained high confidence across both populations were selected. This process guarantees that the recommendations (e.g., *Organic Bananas* -> *Organic Hass Avocado*) are **structurally robust** and likely to convert even for new, unknown customers.
- **Visual Diagnostics:** A **Confidence Stability Plot** was used to identify "Robust Rules" (clustered near the 45-degree diagonal) versus "Overfitted Rules" (which fail to generalize to the test set).



3.3 Feature Engineering

To transition from raw transactional logs to actionable customer segments, we performed extensive feature engineering. We shifted the granularity from "order-level" to "user-level," constructing 11 high-dimensional behavioral proxies.

3.3.1 Economic & Health Signaling (PSI & HAI)

The core of our economic profiling rests on two proprietary indices:

- **Price Sensitivity Index (PSI):** Instead of using absolute prices, we calculated a Relative Price for every product compared to its department average. The PSI represents the percentage of "cheap" products (priced below department mean) in a user's history. This normalizes price sensitivity across categories (e.g., a "cheap" organic meat item vs. a "cheap" snack).
- **Healthy Affinity Index (HAI):** We mapped NutriScore grades (A-E) to a numerical scale (1-5). The HAI measures the proportion of "High-Nutri" (A & B) items in the customer's basket, serving as a proxy for health-conscious shopping behavior.

3.3.2. Diversity and Exploration Metrics

- **Category Entropy:** We implemented a **Shannon Entropy*** calculation on department IDs for each user. A high entropy score indicates a "Generalist" shopper who explores many departments, while low entropy identifies "Specialists" (e.g., users buying almost exclusively in the 'Baby' or 'Pets' departments).
- **Aisle & Department Penetration:** We measured how deeply a user interacts with the store's catalog, providing a metric for "Customer Breadth."

3.3.3 Loyalty & Momentum Features

Beyond standard RFM (Recency, Frequency, Monetary) metrics, we engineered:

- **Order Frequency (Adjusted):** Calculated as total orders divided by average days between orders, providing a velocity metric of the customer's engagement.
- **Basket Volatility:** Using the standard deviation of basket size to distinguish between "Routine Stock-up" shoppers (consistent size) and "Impulse" shoppers (high variance).
- **Reorder Momentum:** A global reorder ratio per user to identify habitual vs. exploratory purchasing patterns.

3.3.4 Product-Level Enrichment

Parallel to customer profiling, we enriched the product catalog with:

- **Product Utility:** The product of average price and total volume, identifying "Cash Cows" for the retailer.
- **Market Penetration:** The percentage of the total user base that has purchased the item at least once, used to distinguish between "Mass Market" and "Niche" anchors.

3.4. Customer Segmentation: An Unsupervised Learning Pipeline

To transform high-dimensional customer features into actionable personas, we implemented a multi-stage clustering pipeline. The methodology was designed to ensure not just statistical separation, but also mathematical stability and business interpretability.

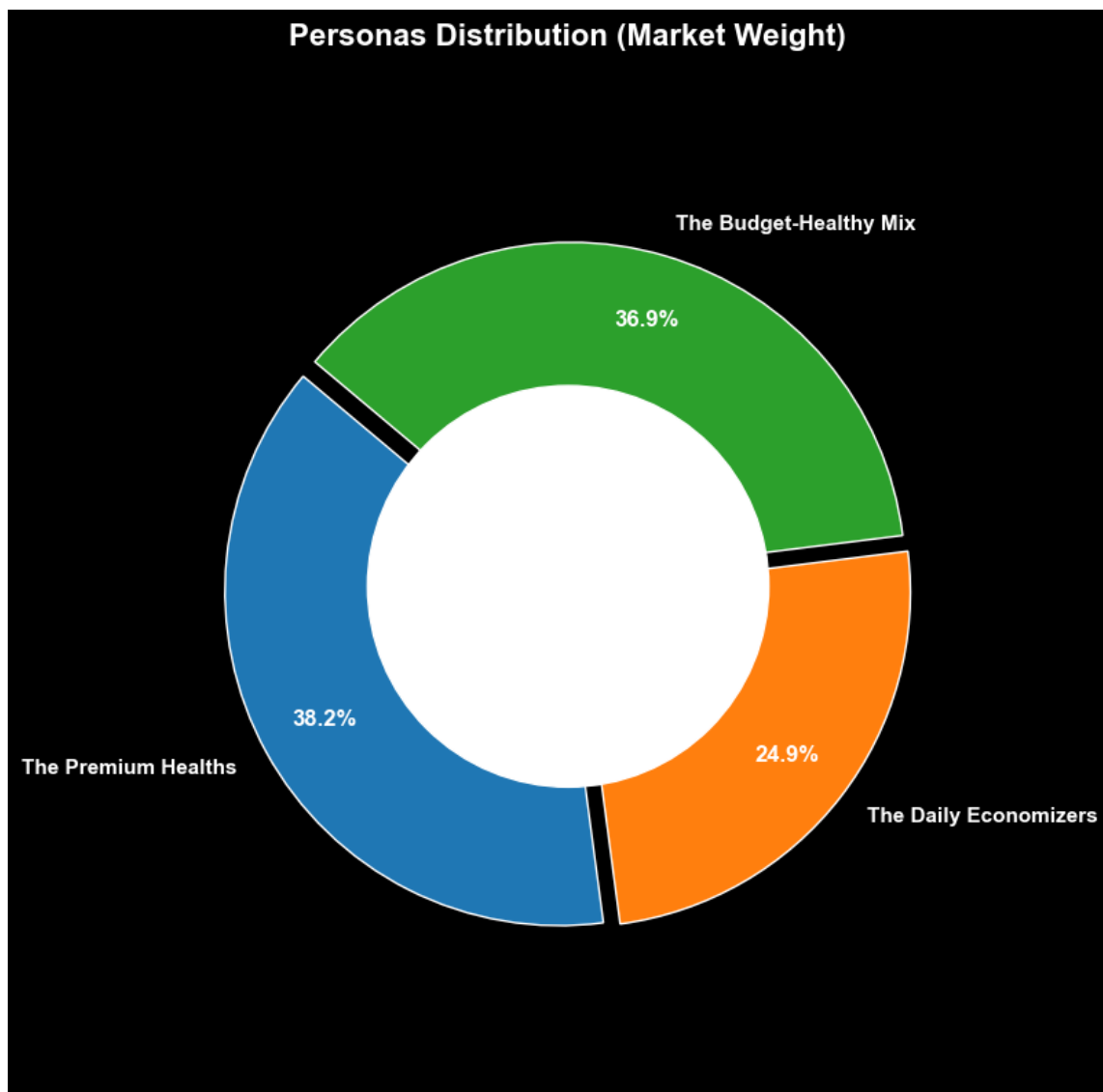
3.4.1. Preprocessing & Dimensionality Alignment

Before clustering, the feature space underwent rigorous preparation:

- **Log-Transformation:** Skewed variables (e.g., *Average Basket Value*) were log-transformed to mitigate the influence of extreme outliers and normalize the distribution.
- **Standardization:** Using `StandardScaler`, all features were centered and scaled to unit variance, ensuring that distance metrics (Euclidean) were not biased by the differing scales of indices (e.g., PSI vs. Basket Size).

3.4.2. Clustering Architecture & Refinement

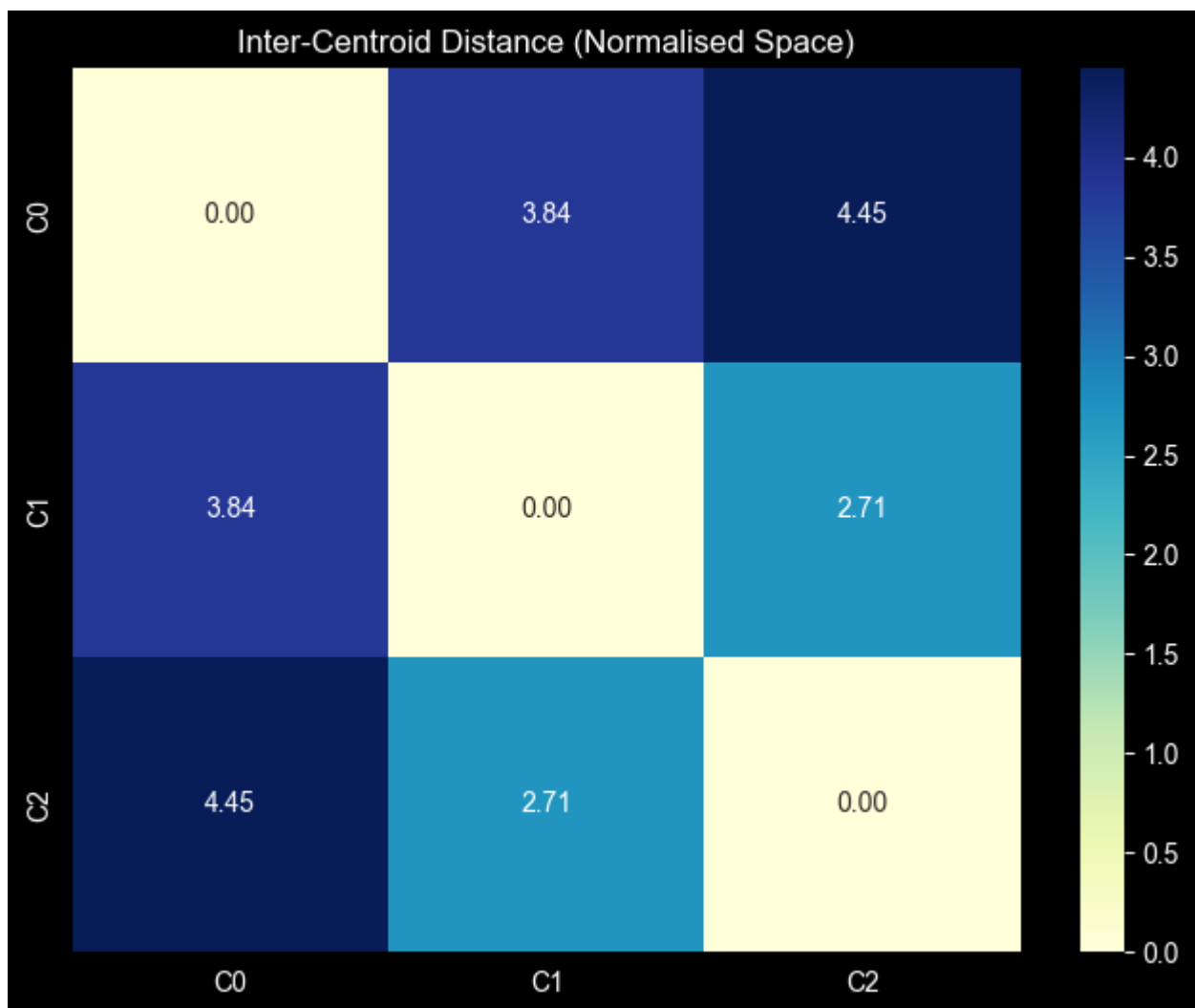
- **Algorithm Selection:** We utilized **K-Means++ initialization** to improve convergence speed and avoid local optima. A **MiniBatchKMeans** approach was initially used for rapid hyperparameter iteration on the 33M-record dataset.
- **Optimal K-Selection:** The "Elbow Method" (Inertia) and **Silhouette Scoring** (estimated on a 60,000-point representative sample) were used to confirm that K=3 provided the optimal balance between granularity and segment stability.
- **Anomaly Detection:** We implemented a distance-based filter to identify **Atypical Customers** (top 1% by distance from centroids), ensuring the final personas were representative of the core customer base.



3.4.3. Validation & Stability Metrics

To ensure the clusters were not artifacts of random initialization, we performed a **Stability Stress Test**:

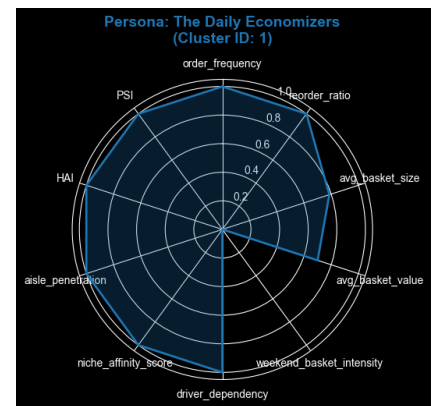
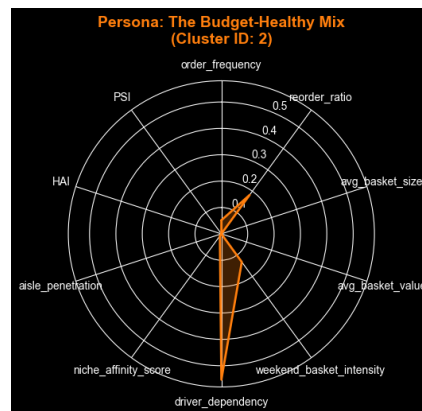
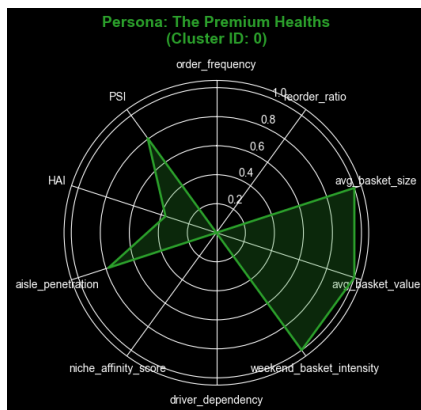
- **Adjusted Rand Index (ARI):** We measured the consistency of cluster assignments across different random seeds. An ARI > 0.85 confirmed a highly stable segmentation.
- **Inter-Centroid Distance:** A heatmap of Euclidean distances in the normalized space was used to ensure significant separation between the three identified cores.



3.4.4. Persona Profiling (The Behavioral Signatures)

The three segments were characterized using Multi-dimensional Radar Charts to visualize their "behavioral signatures":

1. **The Premium Health-Oriented Shoppers:** High HAI (Healthy Affinity), frequent interaction with "Niche" high-fidelity products, and low price sensitivity.
2. **The Daily Economizers:** Highest order frequency and high PSI (Price Sensitivity). These users prioritize routine and cost-efficiency.
3. **The Budget-Healthy Mix:** A hybrid segment showing mid-range health scores and balanced basket sizes, representing the "Value" core of the supermarket.



3.4.5. Model Serialization for Production

The final pipeline was serialized (using joblib), including the trained **StandardScaler** and the **K-Means model**, enabling real-time inference and cluster assignment for new incoming customer data.

3.5. Segment-Specific Strategy: The "Golden Basket" Optimization

To move from behavioural segments to operational action, we developed the **Golden Basket** framework. This system generates a curated assortment of products for each persona, optimized for both customer relevance and retailer objectives.

3.5.1. The Diffusion-Fidelity Matrix (Strategic Characterization)

Instead of ranking products by simple frequency, we categorized the entire catalog (per segment) into five strategic roles using dynamic thresholding (cluster-specific quantiles):

- **Core Products:** High reach and high loyalty. These are the "must-haves" that anchor the customer to the store.
- **Niche Products:** Low reach but extreme loyalty. Critical for long-term retention of specialized shoppers.
- **Opportunistic:** High reach but low loyalty. Products often bought on impulse or during mass promotions.
- **Companion & Fillers:** Secondary products that complete the basket.

3.5.2. Persona-Weighted Scoring (Adaptive Logic)

The "Golden Basket" isn't a static list; it's the result of an adaptive scoring engine that adjusts its priorities based on the targeted Persona:

- **The Premium Healths:** Scoring weight is skewed towards **Healthy Affinity (70%)**, prioritizing products with high NutriScore alignment.
- **The Daily Economizers:** Scoring weight is skewed towards **Price Sensitivity (70%)**, prioritizing products identified as "cheap" relative to their category mean.
- **The Budget-Healthy Mix:** A balanced 40/40/20 weight distribution across price, health, and popularity.

3.5.3. Dynamic Size & Quota Constraints

To ensure the recommendations remain realistic, the size of each Golden Basket is dynamically capped by the segment's **average basket size (+1 item)**. We implemented a **Stratified Selection Logic** to ensure the basket is not just "Core" products, but a balanced mix (e.g., 40% Core, 30% Companion, 20% Niche), preventing recommendation fatigue.

3.5.4. Data Integrity & Coverage

- **NutriScore Layer:** Achieved a basket-level coverage of **27.63%**, providing a statistically significant signal for health-based scoring without restricting the dataset size.
- **Scale:** The enrichment was applied to the full **33.8 million transaction** corpus using **Polars LazyFrames**, ensuring that the resulting "Golden Baskets" are backed by maximum statistical support.

3.6. Predictive Couponing & Campaign Orchestration

To monetize the insights derived from the Golden Baskets and Association Rules, we engineered an **Automated Recommendation Engine**. This system identifies the optimal product, the optimal discount, and—crucially—the optimal timing for each individual customer.

3.6.1. Segmented Market Basket Analysis (SMBA)

Instead of a one-size-fits-all recommendation, we executed the association rule algorithm independently per cluster.

- **Logic:** By calculating Lift and Confidence within each persona, we identify triggers that are specific to shopping styles (e.g., suggesting a specific organic yogurt to a "Premium" shopper vs. a bulk staple to a "Daily Economizer").
- **Optimization:** We utilized Polars LazyFrames to process the 33M rows, applying frequency filters and basket-breadth limits (max 20 items) to ensure statistical significance and computational efficiency.

3.6.2. The "80% Rule" for Predictive Timing

A promotion's ROI depends on its timing. Sending a coupon for milk the day after a customer bought 4 liters is a waste of margin.

- **Repurchase Cycle Analysis:** We calculated the mean days_since_prior_order for every product within each cluster to determine the individual replenishment cycle.
- **Anticipatory Trigger:** The system schedules the promotion to be sent at 80% of the average cycle. This "just-in-time" delivery captures the customer's intent exactly when their stock is running low, but before they have committed to their next shopping trip.

3.6.3. Opportunity-Based Discounting

We moved away from uniform discounts to a **Value-Aware** logic:

- **High-Discovery (20% OFF):** Applied to product pairs with a Lift > 10. These are strong, non-obvious associations. A higher discount is used to incentivize the customer to "break" their routine and discover a new product.
- **Loyalty Maintenance (10% OFF):** Applied to standard replenishment items. Since the probability of purchase is already high (high confidence), a smaller incentive is sufficient to ensure the purchase happens at our store.

3.6.4. Final Campaign Execution

The final output is a unique campaign plan for over **50,000 users**, where each entry includes:

1. **The Trigger:** The recent purchase (from the last 5 items) that justifies the offer.
2. **The Target:** The recommended product with the highest **Opportunity Score** (*Lift x Confidence*).
3. **The Schedule:** The precise number of days to wait before deployment.

3.7. Financial Impact & ROI Analysis: From Data to Profit

The final stage of this project moves beyond predictive accuracy to evaluate **Net Profitability**. The objective is to ensure that the recommendation engine acts as a profit driver rather than a margin-eroding tool.

3.7.1. The "Value-Aware" Profit Formula

To calculate a realistic ROI, we developed a proprietary profit model that integrates financial constraints directly into the algorithm's decision-making process:

- **Departmental Margins:** We applied specific gross margins ranging from 15% (Dairy/Babies) to 40% (Alcohol) to reflect real-world retail economics.
- **The Net Margin Constraint:** A promotion is only validated ("**GO**") if the Net Unit Margin remains positive after deducting the discount cost.
- **Cannibalization Factor (0.7):** We applied a conservative factor to account for the fact that 70% of promoted sales might have occurred anyway. This ensures our profit estimates are not over-optimistic.

3.7.2. Sales Uplift Modeling

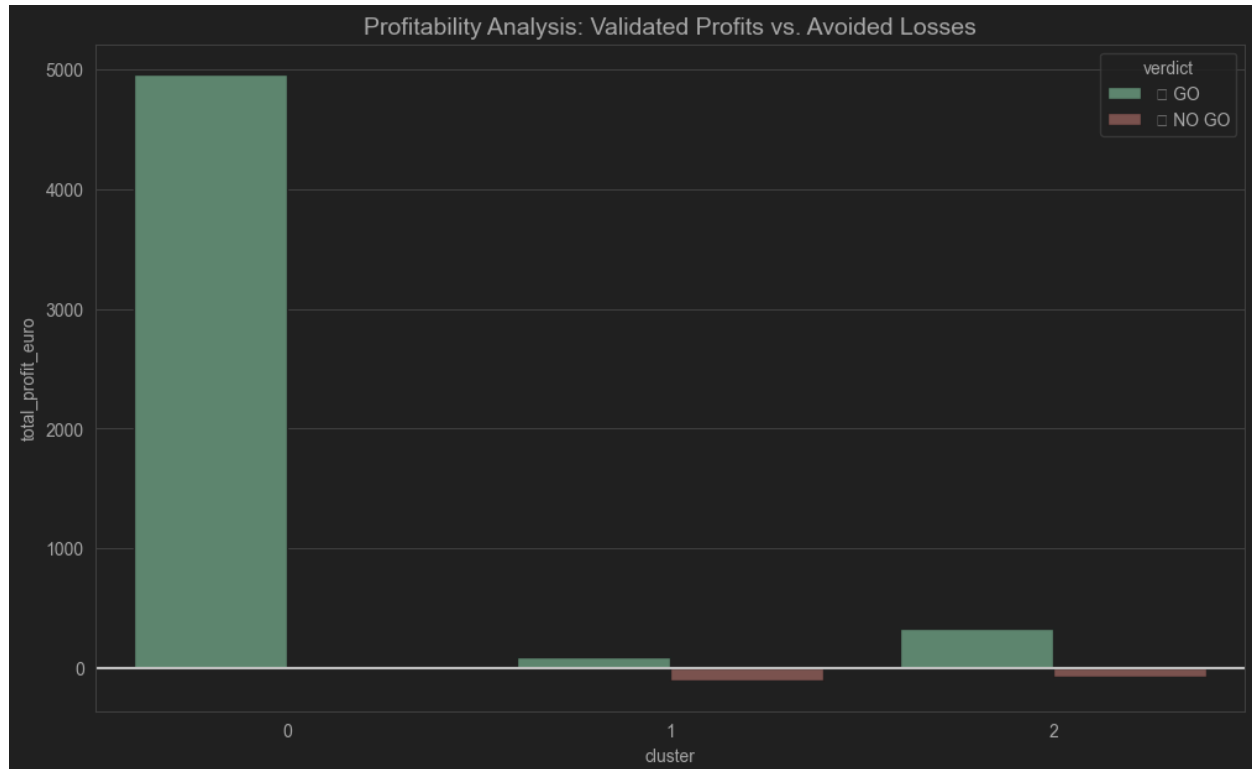
We modeled the impact of discounts on customer behavior using a tiered uplift approach:

- **10% Discount:** Estimated **+15% increase** in purchase probability.
- **20% Discount:** Estimated **+35% increase** in purchase probability (targeted at high-lift "discovery" products).

3.7.3. Performance Results & Margin Protection

The algorithm processed over 50,000 potential offers and generated a strategic verdict for each:

- **Validated Net Profit:** The system identified a net gain of **€5,378.44** from profitable "GO" offers. These represent high-confidence, high-margin opportunities where the promotion effectively drives incremental value.
- **Avoided Losses (Margin Shield):** Crucially, the model identified and blocked **€173.62** in potential losses. These "NO GO" instances represent cases where a standard uniform discount would have cost the retailer more than the product's total margin—a common trap in traditional retail.



3.7.4. Strategic Delivery: The Manager's Decision Tool

The final output of this project is a **Line-by-Line Decision Report**. This tool allows store managers to:

1. Review individual customer offers.
2. Understand the "Why" (Trigger product vs. Recommended product).
3. See the predicted financial outcome before the campaign is even launched.

4. Results & Interpretation

4.1 Customer Segments: Not All Shoppers Are the Same

The 3-cluster solution was validated by a high Adjusted Rand Index ($ARI > 0.80$) and Silhouette scores, ensuring these segments are not random artifacts but stable behavioral structures what many retailers intuitively suspect but rarely quantify: customers do not behave uniformly.

The three segments identified display clear behavioral differences in purchasing intensity, loyalty patterns, and basket economics.

The **Daily Economizers** represent the most active shoppers. They exhibit the highest order frequency and the largest average basket size. Their behavior suggests routine purchasing habits, potentially driven by household demand. Because of their consistent activity, this segment represents a strong opportunity for incremental revenue through strategic bundling.

The **Premium Health-Oriented** has the highest HAI (Healthy Affinity Index) segment purchases less frequently but demonstrates higher alignment with health indicators and stable basket value. These customers are less likely to respond purely to price-driven promotions and may instead respond better to curated, health-aligned bundles.

The **Budget-Healthy Mix** segment displays moderate frequency and smaller baskets, balancing price sensitivity and health orientation. This group may be responsive to value-focused bundles that preserve perceived quality.

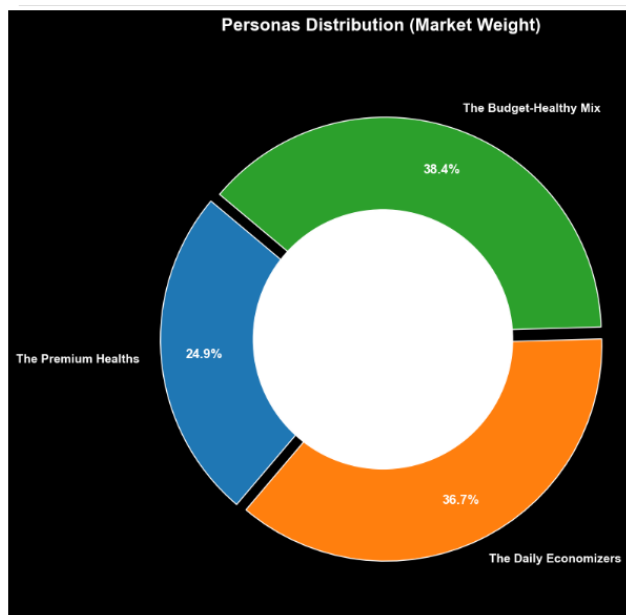


Figure: Behavioral differences across customer segments.

4.2 Association Rules: Understanding Basket Logic

Association rule mining revealed consistent co-purchase patterns within grocery baskets.

Several high-lift rules emerged among fresh produce and organic products, indicating strong complementary demand within that category. For example:

- Customers purchasing organic avocados frequently also purchase bananas.
- Organic strawberries and organic raspberries show strong co-occurrence.
- Lemon purchases are frequently linked with banana purchases.

While these patterns may seem intuitive, their quantified lift values confirm that these relationships are statistically stronger than random co-occurrence.

The FP-Growth algorithm identified 14 stable association rules after applying support and confidence thresholds.

For example:

- {Organic Avocado → Banana} shows a training confidence of 0.303 and a test confidence of 0.297, with a lift of 2.02. This indicates that customers purchasing organic avocado are more than twice as likely to also purchase bananas compared to random expectation.
- {Organic Raspberries → Organic Strawberries} exhibits a lift of 2.96, reflecting particularly strong complementary demand within organic produce.

The minimal difference between training and test confidence values across rules confirms that these patterns generalize beyond the training dataset.

4.3 Frequency Is Not the Same as Value

Introducing pricing data changes the interpretation of frequent bundles. Some of the most common combinations consist of low-priced staples. While statistically strong, promoting these bundles may not meaningfully increase basket value.

In contrast, certain moderately frequent combinations include higher-priced or health-oriented items, contributing more per transaction. This distinction highlights an important insight: frequency describes behavior, but value determines strategic priority.

While staples like Bananas have the highest support, they act as 'Loss Leaders' or 'Traffic Drivers'. Our analysis shifted focus toward 'Value Anchors', products with slightly lower frequency but significantly higher margin contribution.

4.4 The Golden Basket per Segment

The Golden Basket is built on a Diffusion-Fidelity matrix. While 'Core' products ensure basket stability, 'Niche' products (high loyalty, low reach) are the true drivers of segment retention. For the Premium segment, Niche products represent 20% of the Golden Basket, whereas they are secondary for Daily Economizers.

The Golden Basket translates segmentation and association insights into actionable outputs. Products were ranked within each segment using weighted criteria including reach, reorder loyalty, penetration, and estimated spend contribution.

The composition of the Golden Basket differs across personas. Core products dominate all segments, but the balance between companion, niche, and opportunistic items varies. The Daily Economizers show a staple-driven structure, while the Premium Health segment reflects a more selective profile.

These differences justify segment-specific promotions rather than universal bundling.

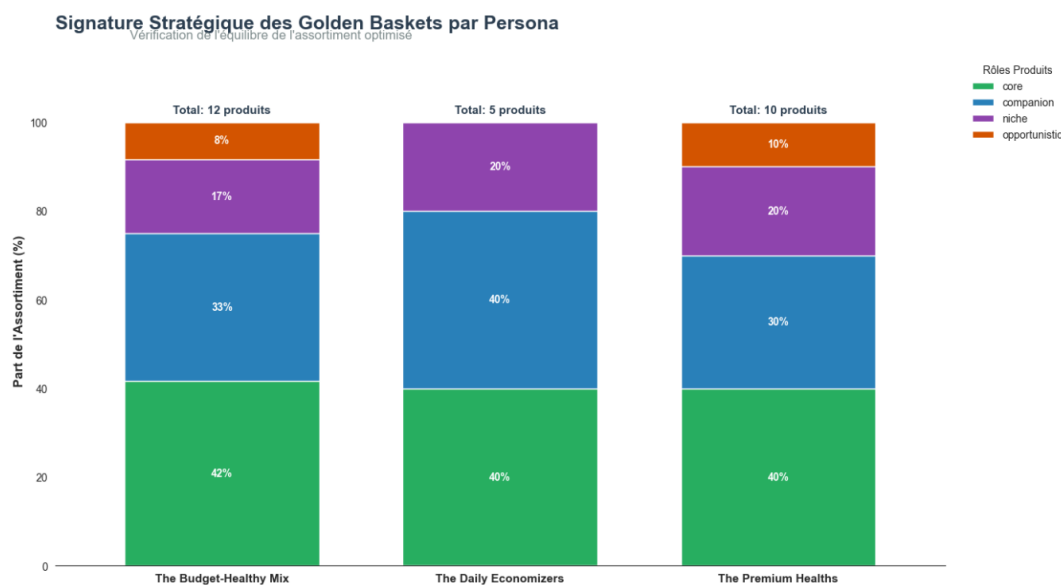


Figure: Strategic Composition of Golden Baskets by Customer Persona

5. Revenue Implications and Business Impact

All previous sections lead to one central question:

What does this mean financially?

The segmentation and Golden Basket analysis demonstrate that customers exhibit structurally different purchasing patterns. Association rules confirm stable co-purchase behavior. Enrichment reveals that not all bundles carry equal monetary weight.

The remaining step is translating these insights into revenue logic.

5.1 From Insight to Incremental Revenue

The Golden Basket per segment provides a shortlist of high-alignment, high-relevance products. These products can be operationalized through:

- Targeted bundle discounts
- Personalized recommendations
- Strategic shelf placement
- Digital promotion campaigns

To estimate potential impact, consider a conservative scenario.

If a retailer targets a specific segment with a Golden Basket bundle and achieves even a modest adoption rate. For example 5% of customers within that segment, incremental revenue can be approximated as:

Projected Uplift = (Segment Size) × (Adoption Rate) × (Incremental Basket Value)

For illustration:

If 5% of a 75,000-customer segment adopt a bundle that increases average basket value by €3, the result is:

$0.05 \times 75,000 \times €3 = €11,250$ per campaign cycle.

Even under cautious assumptions, the impact becomes material at scale.

The objective is not to assume unrealistic conversion rates, but to demonstrate that small behavioral shifts across large customer bases compound into meaningful financial outcomes.

5.2 Reducing Promotional Waste

Revenue growth is only one side of the equation. Cost savings also matter.

Blanket promotions apply discounts to customers who would have purchased the product regardless. This reduces margin without changing behavior.

Segment-specific Golden Baskets reduce this inefficiency by:

- Targeting customers aligned with specific product profiles
- Avoiding unnecessary discounts on low-margin staples
- Prioritizing higher-value combinations

In other words, the framework improves not just revenue generation, but promotional precision.

5.3 Strategic Implications

The key business implications are clear:

1. Customer heterogeneity must inform promotion design.
2. Frequency-based bundling is incomplete without value awareness.
3. Stable association rules provide reliable cross-selling logic.
4. Segment-specific Golden Baskets offer directly operational outputs.

This project demonstrates that transactional data, when structured properly, can shift retail strategy from reactive discounting to targeted revenue optimization.

6. Limitations

While the analytical framework provides meaningful strategic insight, several limitations should be acknowledged.

First, pricing enrichment relies on proxy data rather than retailer-specific margin information. Revenue simulations are therefore indicative rather than exact.

Second, NutriScore coverage is partial. Health-based insights reflect available matches rather than the full product catalog.

Third, clustering in behavioral data rarely produces perfectly separated segments. Although the three-cluster solution is interpretable and actionable, customer behavior remains fluid.

Finally, the analysis is based on historical data and does not account for seasonality or real-time behavioral shifts.

These limitations define the scope of interpretation but do not undermine the structural insights derived from the analysis.

7. Conclusion

This project demonstrates how transactional retail data can be translated into financially actionable strategy.

Through customer segmentation, validated association rule mining, and value-aware enrichment, we moved beyond descriptive frequency analysis toward structured monetization logic.

The findings show that:

- Customer purchasing behavior is heterogeneous.
- Stable co-purchase relationships can be reliably identified.
- Frequency alone does not determine financial importance.
- Segment-specific Golden Baskets provide directly operational outputs.

Rather than relying on uniform promotions, retailers can leverage these insights to design targeted strategies aligned with behavioral structure and economic contribution.

The core contribution of this work is not the identification of patterns, but the translation of those patterns into measurable business logic.